Homework Assignment 4

Due: 2/21

- You must show all intermediate calculations/results.

Problem 1 (10 points). Consider the following dataset:

ID	A1	A2	A3	Class
1	Hot	East	Low	Y
2	Mild	West	High	Y
3	Cool	East	Middle	N
4	Mild	West	Low	Y
5	Hot	East	Middle	Y
6	Cool	West	Middle	N
7	Hot	East	High	Y
8	Mild	West	Low	Y
9	Hot	East	Low	N
10	Cool	West	High	N
11	Mild	West	Middle	Y

Classify a new object O = <A1 = Hot, A2 = West, A3 = Low> using the Naïve Bayes algorithm we discussed in the class.

problem 2 (10 points). Info gain calculation

Consider the following dataset:

ID	A1	A2	Class
1	Hot	East	Y
2	Mild	West	Y
3	Cool	East	N
4	Mild	West	Y
5	Hot	East	Y
6	Cool	West	N
7	Hot	East	Y
8	Mild	West	Y
9	Hot	East	N
10	Cool	West	N
11	Mild	West	Y

(1). Calculate the information gain of attribute A1.
(2). Calculate the information gain of attribute A2.
(3). Which is better as the test attribute at the root level?

Problem 3 (10 points). Use the *Bankruptcy_reduced.csv* dataset and use R for this problem. The attribute *Class* is the class attribute.

(1). Convert the data type of the class attribute to factor.
(2). Split the dataset into training and test sets with the 66%-34% ratio. Make sure that you use a stratified splitting method.
(3). Build a Naïve Bayes model from the training dataset.
(4). Test the model on the test dataset.
(5). In your submission file, include the confusion matrix and the prediction accuracy of each class.
(6). Build a decision tree model using the *C5.0* algorithm from the training dataset.
(7). Test the model on the test dataset.
(8). In your submission file, include confusion matrix and the prediction accuracy of each class.

Problem 4 (10 points). Use the *heart_disease_reduced.csv* dataset for this problem. Use R for questions (1), (2), and (3). For question (4), you must do all calculations yourself.

(1). Build a logistic regression model using **only *resting_bp*, *cholesterol*, and the class attribute**.
(2). Show the intercept and the coefficients of the model.
(3). Classify a new patient X = <resting_bp = 137, cholesterol = 250> and show the probability that the class is 1 and the probability that the class is 2.
(4). Using the intercept and the coefficients of the model, manually calculate the probability that the class of the new patient X is 1. You must show all calculation steps.

Submission:

Name your file *LastName_FirstName*_HW4.doc or *LastName_FirstName*_HW4.pdf. If you have multiple files, then combine all files into a single archive file. Name the archive file as *LastName_FirstName*_HW4.EXT. Here, "EXT" is an appropriate archive file extension (e.g., zip or rar). Upload this archive file to Blackboard.